ENGINEERING PROJECT CASE STUDY · IOT / EMBEDDED SYSTEMS

IoT-Enabled Automated Smart Irrigation System

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Certificate of Merit — Consolation, Genesis of the Trailblazers

UN SDG Goal 2 · Zero Hunger — Sustainable Agriculture & Resource Optimization

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github.com/saimedhas/iot-project

1 EXECUTIVE SUMMARY

⚠ PROBLEM

Traditional irrigation relies on static manual schedules and guesswork — driving massive freshwater depletion, soil degradation, and suboptimal crop yields across farmland.

✓ SOLUTION

An autonomous, sensor-driven **closed-loop irrigation system** that monitors soil & ambient conditions in real time, regulates 5V/12V submersible pumps via motor drivers, and streams live telemetry to a web dashboard over Wi-Fi.

2 HARDWARE STACK & EDGE CIRCUIT

COMPONENT	ROLE IN SYSTEM
Arduino Uno (ATmega328P)	Main controller — runs automated threshold decision logic across the sensing & actuation chain.
ESP8266 Wi-Fi Module	Connectivity module — pushes real-time telemetry to the cloud dashboard.
Sensing Suite	Soil moisture sensor (S:1021 raw), MH-RD raindrop module, DHT11 temp/humidity, soil pH/EC parameters.
Actuation	L298N / relay motor driver switching a DC submersible pump & solenoid valves; 16×2 I2C LCD shows live field readings.

3 WORKFLOW ARCHITECTURE

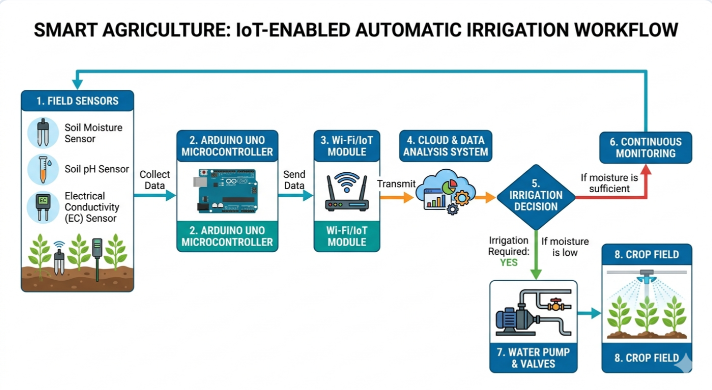

Figure 1 — Smart Agriculture IoT-Enabled Irrigation Workflow (Block Diagram)

4 AUTONOMOUS DECISION ENGINE

A Data Acquisition
Multi-sensor array polls soil moisture, rainfall status, ambient temperature & humidity at calibrated intervals.

B Local Edge Decision
If Soil Moisture > Threshold (dry) AND Rain Status == NO RAIN, the MCU signals the motor driver: Motor = ON.

× Safety Cut-off
Rain detection triggers an immediate override lockout to conserve water during natural precipitation.

D Cloud Sync & Dashboard
Telemetry is sent via ESP8266 over Wi-Fi to a MySQL/cloud backend, updating the live dashboard with timestamps & pump status.

5 45-DAY FIELD EVALUATION

DAY	CONVENTIONAL	IOT SYSTEM
Day 0	0.0	**0.0**
Day 15	0.9	**1.4 (+55%)**
Day 30	1.8	**2.5 (+38%)**
Day 45	2.6	**3.3 (+27%)**

~40% Water conservation vs. manual irrigation

0 Manual interventions for pump regulation

3.3 Crop growth index at Day 45 (IoT)

6 PROTOTYPE & LIVE TELEMETRY

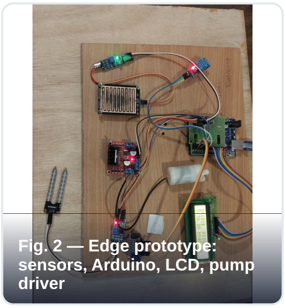

Fig. 2 — Edge prototype: sensors, Arduino, LCD, pump driver

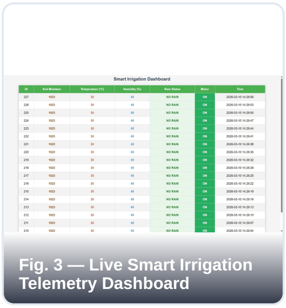

Fig. 3 — Live Smart Irrigation Telemetry Dashboard

7 ENGINEERING HIGHLIGHTS

01 Closed-loop hardware-in-the-loop control connecting microcontrollers, sensor arrays, and high-power inductive pump loads.

02 Resilient Wi-Fi data pipelining with cloud logging (ESP8266 → MySQL) for automated environmental telemetry.

03 Dual feedback architecture — real-time on-field LCD display plus a centralized, remotely accessible web dashboard.